

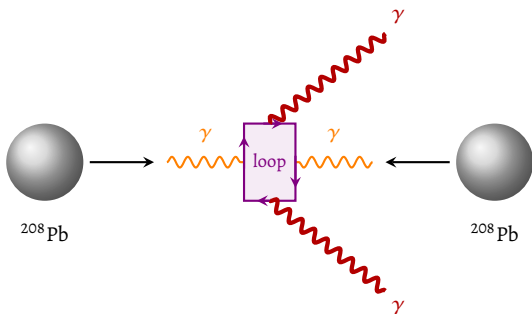


Figure 5 – Light-by-light scattering ($\gamma\gamma \rightarrow \gamma\gamma$) in ultra-peripheral Pb–Pb collisions at the LHC. Each relativistic nucleus emits a quasi-real photon; these interact via a loop of virtual fermions (*box* diagram), producing two real photons in the final state ([ATLAS Collaboration, 2017](#); [ATLAS Collaboration, 2019](#)).

